Ghana's Digital Economy: A Comparative Analysis

World Development Indicators (WDI) Capstone Project

Prepared by Richard Courage Cobbinah | AnalystLab Africa, Batch B | July 2026

Objective and Problem Statement

This project measures Ghana's digital economy against five comparator countries: Nigeria, Kenya, Cote d'Ivoire, Senegal, and South Africa, plus Sub-Saharan Africa and World benchmarks. Ghana's banking sector has adopted mobile money at scale. This report checks whether that adoption carries through to broader digital infrastructure and digital services trade, and identifies where Ghana leads or lags its regional peers.

Dataset Description

Source: World Bank World Development Indicators (WDI), bulk CSV download. The full file contains 1,400+ indicators for 217 countries and territories from 1960 to 2025. This analysis draws 10 indicators across three areas:

- Connectivity: Individuals using the Internet (%), Mobile cellular subscriptions (per 100), Fixed broadband subscriptions (per 100), Secure Internet servers (per million)
- Digital finance: Account ownership at a financial institution or mobile money service (% age 15+)
- Digital trade and context: ICT service exports (% of service exports), GDP per capita, total population

Analysis window: 2000 to 2024. 2025 was dropped because most indicators have no reported values for that year.

Data Cleaning Process

All cleaning was performed in Power BI's Power Query Editor, working directly from the WDI master CSV, no external scripting was used.

- Filtered the 398,469-row master file down to the 10 target indicator codes using a custom M filter, cutting the working table to roughly 2,600 rows before reshaping.
- Unpivoted the year columns (wide format, one column per year) into a long format with Year and Value columns, producing tens of thousands of country-indicator-year observations.
- Set correct data types: Year as whole number, Value as decimal number.
- Removed rows with blank/null values, then filtered the Year column to the 2000-2024 analysis window.
- Merged in Region and Income Group by joining against the WDI country metadata file on Country Code, so the dashboard's map and slicers can group and filter correctly.
- Added a calculated IsComparator column (Yes/No) to flag Ghana and its 5 peer countries, used across every chart to scope the comparison set.
- Renamed each indicator to a short, dashboard-friendly label using Replace Values (e.g. "Individuals using the Internet (% of population)" became "Internet Users (%)").
- Checked coverage per country per indicator. All comparator countries had 15 or more years of data per indicator across 2000 to 2024.
- Flagged one data quality note: Ghana's mobile cellular subscriptions series shows a drop in 2019 (43.0 to 22.1 per 100 people) before recovering in 2020. This reflects a reporting or methodology change in the source data, not an actual service disruption. The dashboard keeps this value visible with a note rather

than removing it.

Analysis Methodology

• Trend analysis: tracked each indicator for Ghana from 2000 to 2024 to measure the pace of adoption.

• Peer benchmarking: compared Ghana's latest available value against 5 regional peers, Sub-Saharan Africa, and the World average for each indicator.

• Gap analysis: identified indicators where Ghana ranks near the top of its peer group versus indicators where it ranks near the bottom, to separate genuine strengths from real gaps.

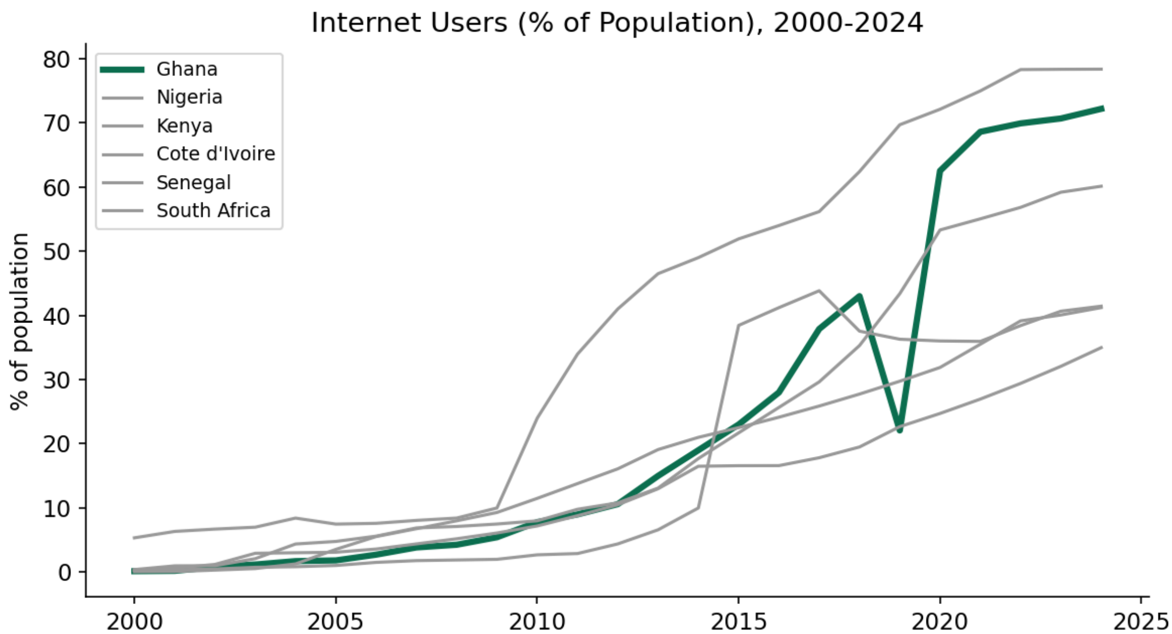
• Correlation analysis: tested whether fixed broadband access and internet use move together across the comparator group, using Pearson's correlation coefficient, to check whether broadband investment is a plausible lever for raising internet use, or whether the two are already decoupled.

• Economic context: layered in GDP per capita trends to check whether Ghana's digital indicators are in line with its income level, ahead of it, or behind it, relative to peers.

Key Findings

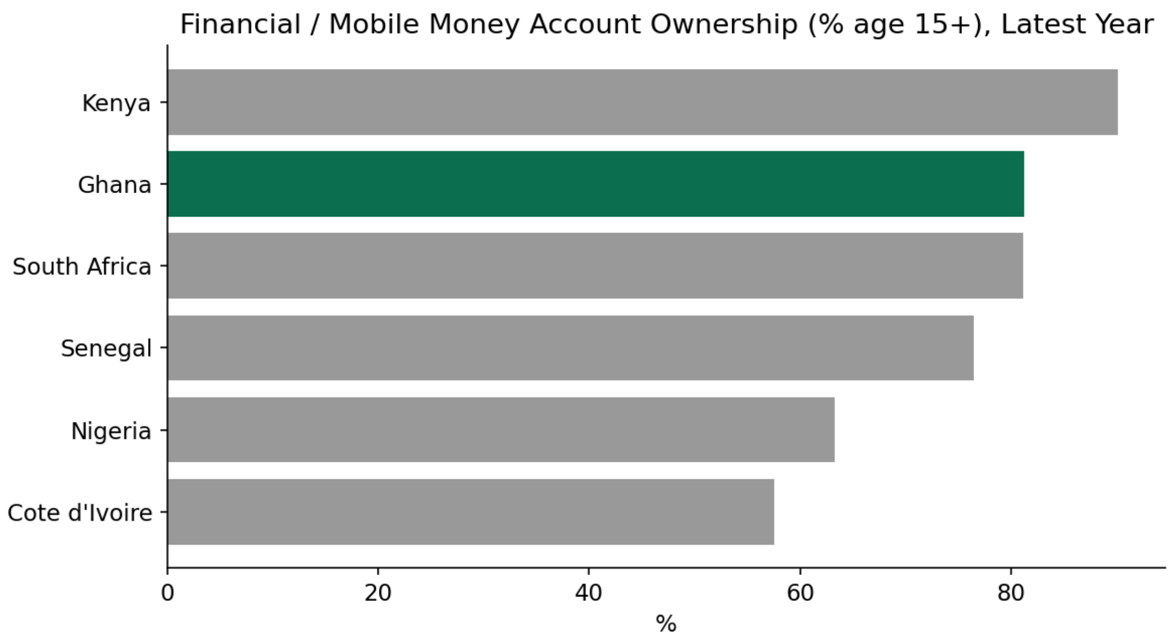
1. Internet adoption in Ghana now leads the comparator group

Ghana reached 72.2% internet penetration in 2024, ahead of South Africa (78.4%) only narrowly, and well above Nigeria (41.2%), Kenya (35.0%), and the Sub-Saharan Africa average (33.6%). Ghana's growth was near zero before 2007, then accelerated steadily to 2018, and jumped again from 2020 onward.



2. Mobile money adoption is Ghana's strongest indicator

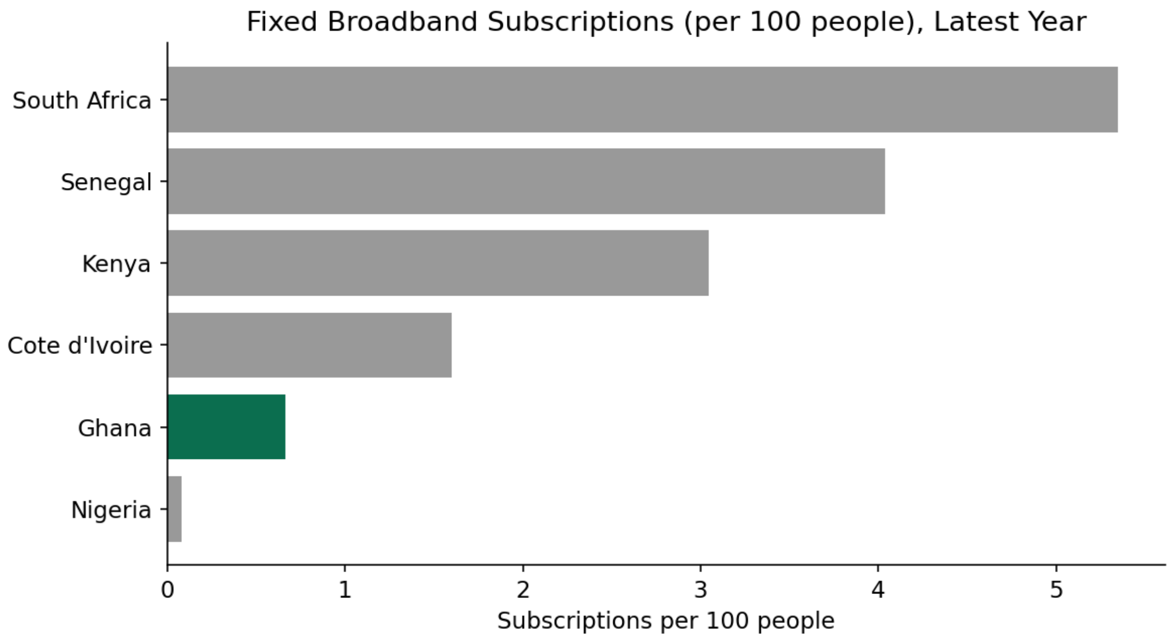
81.2% of Ghanaians aged 15+ own a financial institution or mobile money account, the second-highest rate in the comparator group after Kenya (90.1%), and above South Africa, Senegal, Nigeria, and Cote d'Ivoire. This confirms Ghana's mobile money infrastructure, not traditional banking, is the main driver of financial inclusion.



Financial / Mobile Money Account Ownership (% age 15+), Latest Year
Kenya
Ghana
South Africa
Senegal
Nigeria
Cote d'Ivoire
0
20
40
60
80
%

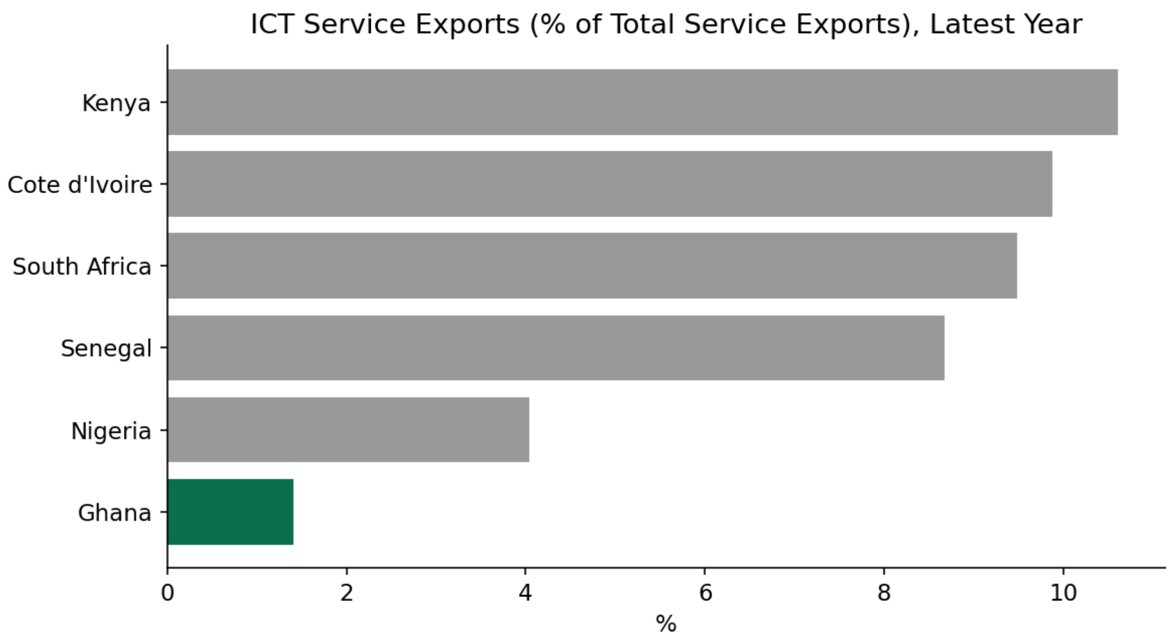
3. Fixed broadband infrastructure is Ghana's weakest indicator

Ghana has 0.66 fixed broadband subscriptions per 100 people, below Nigeria's peer group average and far below South Africa (5.35) and Senegal (4.04). High mobile internet use is compensating for a near-absent fixed broadband market. This limits the kind of high-bandwidth digital services (cloud computing, remote work infrastructure, streaming, data centers) that fixed broadband supports.



4. Ghana exports almost no digital services

ICT service exports make up just 1.4% of Ghana's total service exports, the lowest in the comparator group. Kenya leads at 10.6%, followed by Cote d'Ivoire and South Africa near 9-10%. Ghana has strong digital consumption (internet and mobile money) but has not converted that base into a digital services export industry.

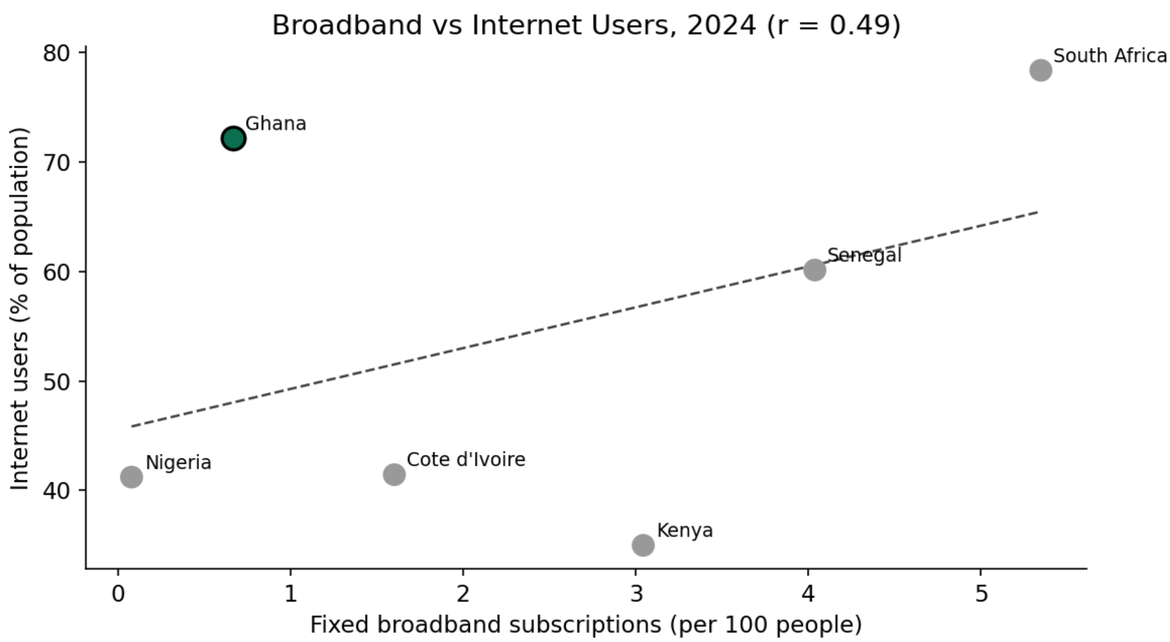

ICT Service Exports (% of Total Service Exports), Latest Year
Kenya
Cote d'Ivoire
South Africa
Senegal
Nigeria
Ghana
0
2
4
6
8
10
%

Extended Analysis

The dashboard goes beyond the four core comparisons above with four additional analyses: a correlation test between broadband and internet use, a full ICT export trend, a GDP per capita trend, and a filled world map for global context.

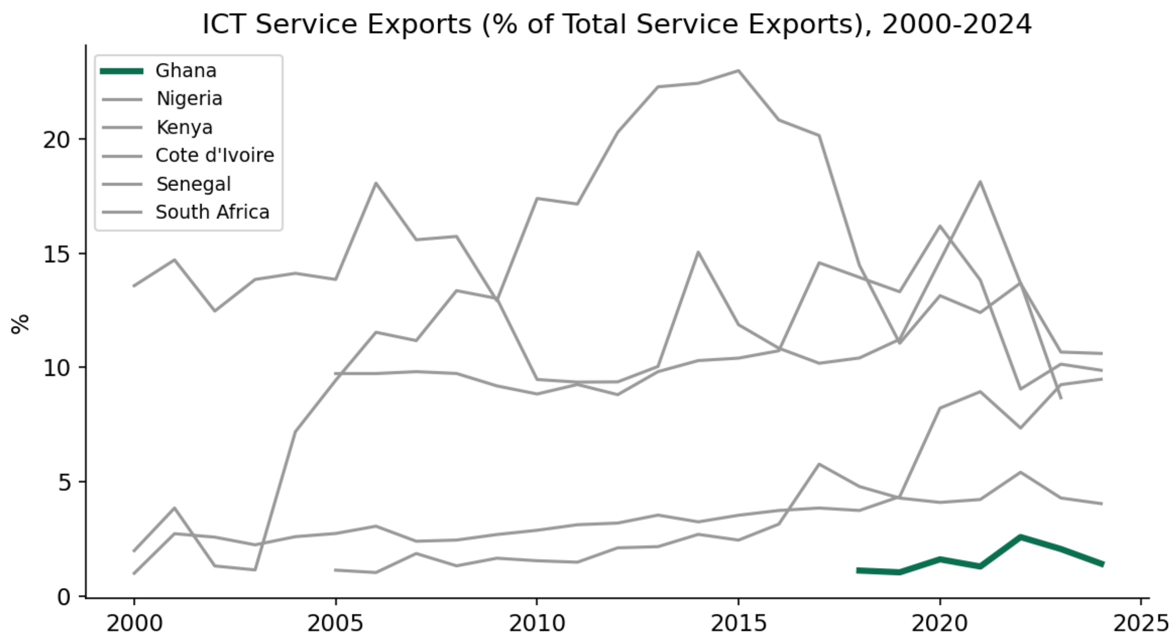
5. Broadband and internet use are only moderately correlated

Across the 6 comparator countries, the Pearson correlation coefficient between fixed broadband subscriptions and internet users (2024) is 0.49. This falls in the moderate range (0.3 to 0.7): broadband access is associated with higher internet use, but the relationship is not strong enough to treat broadband as the primary driver. Ghana is a clear outlier on this trend line: it has the second-lowest broadband subscription rate in the group, yet the second-highest internet penetration, which confirms mobile networks, not fixed broadband, explain Ghana's internet growth.



6. Ghana's ICT export gap is not recent, it is structural

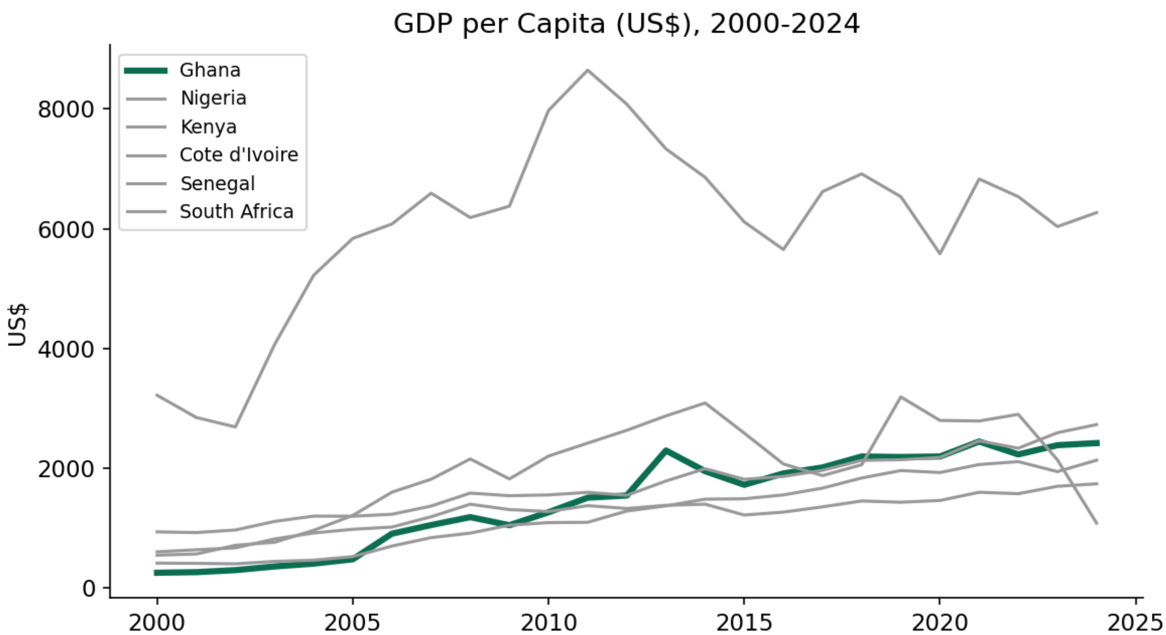
The full 2000-2024 trend shows Ghana's ICT service exports have stayed near the bottom of the comparator group for the entire period, not just in the most recent year. Kenya and Cote d'Ivoire show visible growth phases; Ghana's line stays flat and low throughout. This confirms the export gap identified in the snapshot view is a long-standing structural issue, not a one-year dip, which strengthens the case for a dedicated policy intervention rather than a wait-and-see approach.



ICT Service Exports (% of Total Service Exports), 2000-2024
Ghana
Nigeria
Kenya
Cote d'Ivoire
Senegal
South Africa
%
0
5
10
15
20
2000
2005
2010
2015
2020
2025

7. Ghana's digital gains outpace its income growth

GDP per capita trends show Ghana growing steadily but modestly over the period, tracking closer to Cote d'Ivoire and Kenya than to South Africa. Yet Ghana's internet and mobile money indicators rank at or near the top of the comparator group. This is a notable finding: Ghana's digital economy indicators are running ahead of what its income level alone would predict, most likely a result of mobile infrastructure and financial inclusion policy outperforming general economic growth.



8. Global map view

The dashboard includes a filled world map of internet users by country (2024), giving reviewers global context beyond the 6-country comparator set. Ghana sits visibly above the regional average shade, reinforcing its position as a regional leader on this metric even against a wider field.

Insights

• Ghana's digital economy is mobile-first. Growth in internet use and financial inclusion both ride on mobile networks, not fixed infrastructure, confirmed by the moderate (0.49) broadband-internet correlation and Ghana's outlier position on that trend line.

• The gap between high mobile adoption and near-zero digital exports signals a missed monetization opportunity. Digital consumption has not translated into a digital production or export sector, and the 2000-2024 trend shows this gap has persisted for the full period, not just recently.

• Weak fixed broadband is a structural constraint. It caps the growth of data-heavy digital services and business process outsourcing that require stable, high-bandwidth connections.

• Ghana's digital indicators outperform what its GDP per capita trend alone would predict, evidence that mobile and financial-inclusion policy is doing more work than income growth in driving these outcomes.

• Ghana's 2019 mobile subscription data dip is a reporting artifact worth flagging to any analyst reusing this dataset, not a real market event.

Recommendations

• Investment priority: target fixed broadband infrastructure expansion, particularly in underserved regions outside Accra and Kumasi, to close the widest gap versus peer countries.

• Policy priority: build export incentives and skills pipelines for ICT and business process outsourcing services, using Kenya's model as a reference point.

• Private sector opportunity: Ghana's high mobile money penetration is a ready base for fintech, e-commerce, and digital lending products that build directly on existing mobile infrastructure.

• Data quality action: any organization using WDI mobile subscription data for Ghana should treat the 2019 value as a reporting break, not a trend signal.

Power BI Dashboard Features

The accompanying Power BI dashboard (WDI_Digital_Economy_Ghana.pbix) implements the full analysis above as an interactive report.

KPI cards

Five cards across the top: Internet Users, Broadband, Mobile Money, ICT Exports (all Ghana, 2024), and the Broadband-Internet Correlation coefficient.

Charts

• Internet Users trend line, 2000-2024, all comparator countries

• Fixed Broadband Subscriptions, bar chart, 2024 snapshot

• Financial/Mobile Money Account Ownership, bar chart, 2024 snapshot

• ICT Service Exports, bar chart, 2024 snapshot

• Broadband vs Internet Users, scatter plot with trend line

• ICT Service Exports trend, 2000-2024

- GDP per Capita trend, 2000-2024
- Filled world map, Internet Users by country, 2024, global context

Slicers and interaction design

The dashboard includes Year, Region, and Income Group slicers. Year filters every visual, since every comparator country has data across the full 2000-2024 window. Region and Income Group are scoped to the world map only: Ghana's own comparator group spans a single region (Sub-Saharan Africa) and income band (Lower-middle income), so a slicer that could exclude Ghana from its own dashboard would work against the report's purpose. Region and Income Group instead let a reviewer explore the map's global view without disrupting the Ghana-centered charts and KPIs. This was a deliberate interaction design choice, set using Power BI's Edit Interactions feature, not a limitation of the data.

Conclusion

Ghana leads its regional peer group on internet use and ranks second on mobile money adoption. Both strengths run through mobile networks, not fixed infrastructure, a pattern confirmed by the moderate broadband-internet correlation (0.49) and by Ghana's outlier position on that relationship. Ghana's digital indicators also outperform what its GDP per capita trend would predict, suggesting policy and mobile infrastructure are doing more work than income growth alone. Fixed broadband and digital services exports are the two clear, long-standing gaps, confirmed by the full 2000-2024 trend rather than a single-year snapshot, and closing them is the next stage of Ghana's digital economy growth. The comparator data shows this gap is closable: Kenya and South Africa demonstrate that stronger fixed infrastructure and export-oriented digital services policy produce measurably better outcomes on exactly these indicators.